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CO1 AT01

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

```
# Given Probabilities for selecting sections
prob_A = 0.25  # 1/4
prob_B = 0.50  # 1/2
prob_C = 0.25  # 1/4

# Conditional probabilities of drawing a blue car given the section
prob_blue_given_A = 3 / 10
prob_blue_given_B = 6 / 15
prob_blue_given_C = 2 / 15

# Law of Total Probability to find P(Blue)
total_prob_blue = (prob_A * prob_blue_given_A) + \
                  (prob_B * prob_blue_given_B) + \
                  (prob_C * prob_blue_given_C)

# Applying Bayes' Formula for P(Section C | Blue)
prob_C_given_blue = (prob_C * prob_blue_given_C) / total_prob_blue

print(f"The probability that the blue car originated from Section C is:
{prob_C_given_blue:.4f}")
print(f"In percentage: {prob_C_given_blue * 100:.2f}%")
```

```
The probability that the blue car originated from Section C is: 0.1081
In percentage: 10.81%
```

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4

(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2

(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0

(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1

(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3

(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6

(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9

(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

> ***Answers:***
> *i) The model begins to overfit at* ***Epoch 4****. At this point, the training loss continues to drop, but the validation loss reverses its trend and starts increasing.*
> *ii) Two highly effective techniques are* ***Early Stopping*** *(halting the training process when validation loss stops improving) and* ***Dropout*** *(randomly disabling a percentage of neurons during training).*
> *iii) Overfitting severely hinders generalization. The model essentially "memorizes" the specific noise of the training data, making it highly inaccurate and unreliable when processing new, unseen real-world data.*

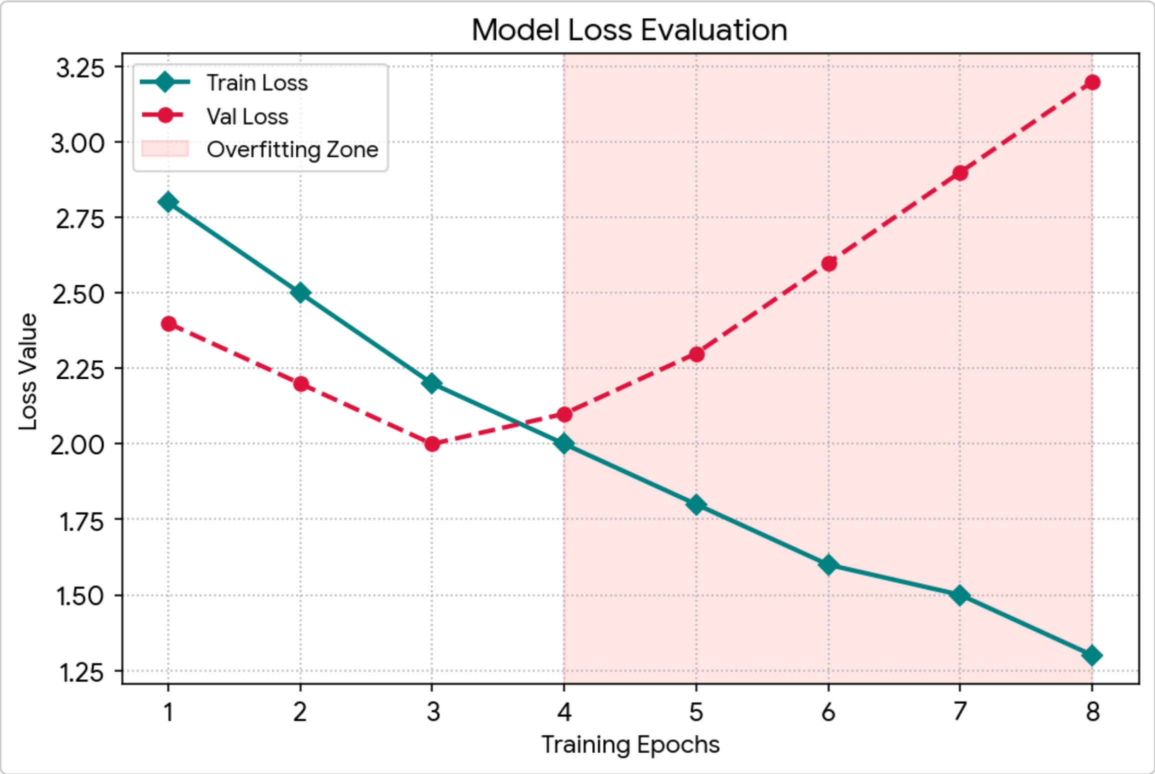
```
import matplotlib.pyplot as plt

epochs = [1, 2, 3, 4, 5, 6, 7, 8]
train_loss = [2.8, 2.5, 2.2, 2.0, 1.8, 1.6, 1.5, 1.3]
val_loss = [2.4, 2.2, 2.0, 2.1, 2.3, 2.6, 2.9, 3.2]

plt.figure(figsize=(8, 5))
plt.plot(epochs, train_loss, color='teal', linewidth=2, marker='D', label='Train
Loss')
plt.plot(epochs, val_loss, color='crimson', linewidth=2, linestyle='--', marker='o',
label='Val Loss')

# Highlight overfitting with a shaded region
plt.axvspan(4, 8, color='red', alpha=0.1, label='Overfitting Zone')

plt.title("Model Loss Evaluation")
plt.xlabel("Training Epochs")
plt.ylabel("Loss Value")
plt.legend()
plt.show()
```



3. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

(h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

> ***Answers:***
> *i) Overfitting clearly begins at **Epoch 6**, where validation accuracy peaks at Epoch 5 (73%) and then drops to 72%.*
> *ii) Validation accuracy drops because the network becomes overly complex and too highly tuned to the specific examples in the training set, thereby losing its ability to generalize to the validation set.*
> *iii) To improve validation performance, you can implement **Data Augmentation** (artificially increasing the diversity of the training set) or apply **L2 Regularization (Weight Decay)** to penalize large weights.*

```
import matplotlib.pyplot as plt

epochs = [1, 2, 3, 4, 5, 6, 7, 8]
train_acc = [60, 65, 70, 75, 80, 85, 88, 92]
val_acc = [58, 63, 68, 72, 73, 72, 70, 68]

plt.figure(figsize=(8, 5))
plt.plot(epochs, train_acc, color='navy', linewidth=2, marker='^', label='Train
Accuracy')
plt.plot(epochs, val_acc, color='darkorange', linewidth=2, linestyle='-.', marker='s',
label='Val Accuracy')

# Highlight overfitting with a shaded region
plt.axvspan(6, 8, color='orange', alpha=0.15, label='Overfitting Zone')

plt.title("Model Accuracy Evaluation")
plt.xlabel("Training Epochs")
plt.ylabel("Accuracy (%)")
plt.legend()
plt.show()
```